

# Toward Visually Robust Quadruped Parkour via Monocular Depth Estimation

Background, methodology, and experiments on monocular depth for parkour

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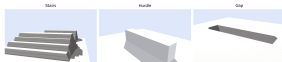
# Problem

# What Is Quadruped Parkour?

## Working Definition

Quadruped parkour is **dynamic locomotion over discrete obstacles** such as stairs, hurdles, and gaps, where success depends on

- perceiving terrain geometry,
- choosing the right footholds and timing,
- staying stable through contact changes.



## Course or robot visual

Insert a simple schematic or photo sequence of a quadruped approaching, crossing, and landing over one obstacle.

Obstacle families used in the thesis: stairs, hurdles, and gaps.

# Motivation and Thesis Question

- Most visual parkour systems deploy with an onboard **depth camera**.
- RGB is easier to deploy, but the visual sim-to-real gap is harder.
- I studied whether **monocular depth** can act as an intermediate representation.

## Thesis Question

Can a pretrained monocular depth estimator preserve enough terrain geometry for a distilled quadruped parkour policy to act reliably?

## Contribution Style

This thesis is mainly an **experimental and methodological study**: what I tried, what worked, what failed, and what that says about the role of perception.

**Background**

# Key Development 1: From Agile Locomotion to Perceptive Locomotion

## Hwangbo et al. (2019)

Learned agile motor skills for quadrupeds in simulation and transferred them to hardware.

## Lee et al. (2020)

Pushed learned locomotion onto challenging terrain using privileged training ideas.

## Miki et al. (2022)

Made locomotion **perceptive**: onboard depth sensing became part of the control loop.

### Visual strip from the foundation papers

Suggested visuals: a dynamic ANYmal sequence from Hwangbo 2019, a challenging-terrain figure from Lee 2020, and a depth-camera perceptive locomotion visual from Miki 2022.

Key foundations cited in the thesis related-work section.

## Key Development 2: Visual Parkour Systems

Agarwal et al. (2023)

Introduced egocentric vision with **scandots** and a teacher–student pipeline for difficult terrain.

Zhuang (2023)  
Cheng (2024)

Robot Parkour Learning and Extreme Parkour pushed to fast, highly visual obstacle traversal on real quadrupeds.

Hoeller (2024)  
Chane-Sane (2025)

ANYmal Parkour and SoloParkour explored hierarchical control and privileged-experience training.

### Parkour visual from prior work

Suggested visual: a real-robot obstacle-course image or motion sequence from Cheng 2024 or Zhuang 2023.

### Representation / architecture visual

Suggested visual: scandot representation or teacher–student pipeline from Agarwal 2023.

# Key Development 3: RGB Sim-to-Real and Monocular Depth

## RGB Transfer Strategies

- **Domain randomization:** Tobin et al. (2017)
- **RGB parkour:** Jenkins (2024)
- **Generated images:** Yu et al. (LucidSim, 2025)

### RGB sim-to-real visual

Suggested visual: domain-randomized vs generated parkour scenes from Jenkins 2024 or LucidSim.

## Monocular Depth Becomes Viable

- MiDaS / zero-shot depth transfer: Ranftl et al. (2022)
- Large pretrained depth foundation models: Yang et al. (DA2, 2024)



My thesis sits at the intersection of parkour control, RGB transfer, and monocular depth estimation.



# Where This Thesis Sits

## Parkour

Strong control pipelines already exist, but they usually assume rich terrain sensing.

## RGB Deployment

RGB is attractive in practice, but direct visual transfer is still brittle.

## Monocular Depth

Foundation models offer a geometry-oriented representation without training a parkour-specific visual encoder from scratch.

## Thesis Positioning

I study whether **monocular depth + teacher-student parkour** is a useful experimental direction, and where that idea breaks.

# Method

# Mathematical Model of the Task

## POMDP

$$(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma), \quad J(\pi) = \mathbb{E}_{\pi} \left[ \sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]$$

with actions  $a_t \in \mathbb{R}^{12}$ .

## Observation Split

$$o_t = (o_{\text{prop}}, o_{\text{ext}})$$

Proprioception carries robot state; exteroception carries terrain geometry.

## Teacher

$$o_t^T = (o_{\text{prop}}, \text{scandots}, \text{priv. params})$$

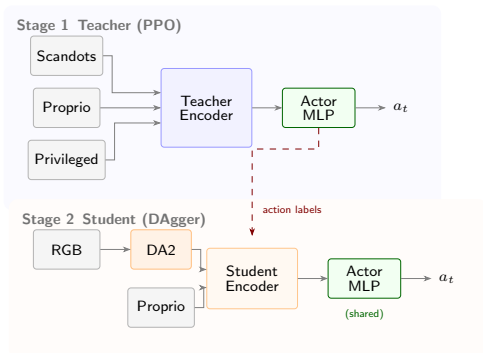
Direct terrain geometry during training.

## Student

$$o_t^S = (o_{\text{prop}}, \text{depth image})$$

Partial observation, so a recurrent encoder is justified.

# Two-Stage Pipeline



- **Teacher:** PPO with privileged geometry.
- **Student:** DAgger from the student's own state distribution.
- **Question:** replace simulator depth with DA2-estimated monocular depth.

## Main idea

First solve control, then test whether perception preserves the geometry the controller needs.

Teacher-student pipeline adapted from the thesis; PPO after Schulman et al. (2017), DAgger after Ross et al. (2011).

# Depth Signal Quality (DSQ)

## Definition

$$\text{DSQ} = \frac{1}{T} \sum_{t=1}^T \frac{1}{N} \sum_{i=1}^N \rho(d_i^{\text{GT}}, d_i^{\text{DA2}})$$

where  $\rho$  is Pearson correlation over pixels.

## Why this metric?

Pearson correlation is invariant to positive affine transforms, so DSQ ignores the scale-shift ambiguity of monocular depth and focuses on **preserved geometry**.

## DSQ explainer visual

Suggested visual: side-by-side depth frames labeled “same shape, different scale” versus “geometry collapsed”.

# Results

# What I Tested

## 1. Baselines

Can the teacher and a GT-depth student solve the course?

## 2. Estimated Depth

What changes when ground-truth depth is replaced by DA2 depth?

## 3. Sensitivity

How robust is the visual pipeline to texture, lighting, and color changes?

### Experiment layout visual

Suggested visual: one slide-friendly schematic showing teacher baseline, GT-depth student, DA2 student, and test-time perturbations.

# Main Findings

## Baseline message

Teacher and GT-depth student are strong; zero-depth almost fails completely.

## Estimated-depth message

Larger DA2 models help, and texture helps the hardest obstacle class: **gaps**.

## Takeaway

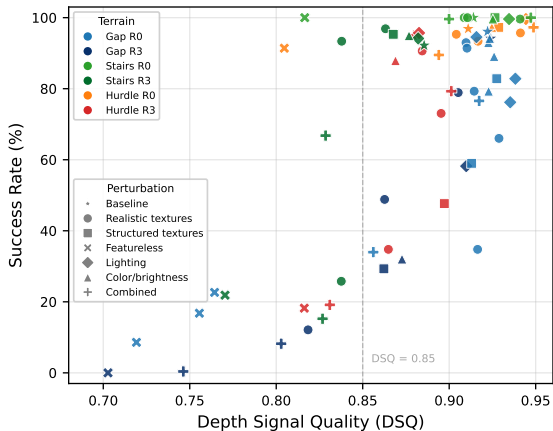
The main bottleneck is **perception quality**, not the downstream controller.

## Simplified comparison chart

Suggested visual: a bar chart comparing GT-depth, DA2-base, and DA2-base + texture on the hardest gap row. Key values from the thesis: GT-depth 87.5%, DA2-base 0.8%, DA2-base + texture 44.9%.



# Sensitivity and the Role of DSQ



DSQ-vs-success scatter computed over multiple obstacle settings and test-time perturbations.

## Observed pattern

- Lighting and color changes are often tolerable.
- Featureless surfaces cause the largest failures.
- Low DSQ usually coincides with failure.

## Interpretation

DSQ is a useful **diagnostic**, but high DSQ alone does not guarantee successful control.

# Takeaways and Limits

## What I learned

- Monocular depth can support nontrivial parkour **in simulation**.
- Teacher–student training is a useful way to isolate perception from control.
- Geometry preservation matters, and DSQ captures part of that story.

## What remains open

- Real-world transfer is untested.
- Texture-poor surfaces are still a major weakness.
- The current depth stack is too heavy for onboard deployment.

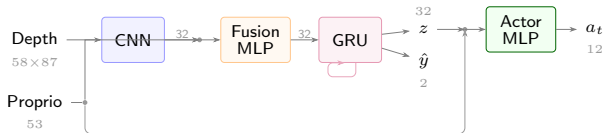
## Bottom line

This thesis suggests that monocular depth is a plausible experimental direction for parkour, but perception quality remains the dominant bottleneck.

**Thank you!**

**Questions?**

## Backup: Student Architecture



- CNN encodes the  $58 \times 87$  depth image.
- Proprioception is fused before the GRU.
- The GRU outputs a terrain latent  $z \in \mathbb{R}^{32}$  and heading estimate  $\hat{y} \in \mathbb{R}^2$ .
- The teacher's actor MLP is reused downstream.

# Backup: Reward and Training Details

## Teacher Reward Structure

- Task progress and success bonuses
- Termination and collision penalties
- Torque, power, and smoothness regularization
- Orientation and body-stability penalties

## Control

12 actuated joints, 50 Hz action rate, PD controller with  $K_p = 30.0$  and  $K_d = 0.75$ .

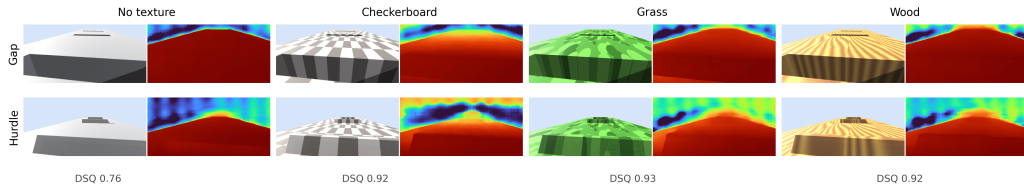
## Key PPO Hyperparameters

|                 |           |
|-----------------|-----------|
| Environments    | 4096      |
| LR              | $10^{-3}$ |
| $\gamma$        | 0.99      |
| $\lambda$       | 0.95      |
| Clip $\epsilon$ | 0.2       |

## Key DAgger Hyperparameters

|                |                    |
|----------------|--------------------|
| Environments   | 48                 |
| BPTT window    | 24                 |
| LR             | $2 \times 10^{-3}$ |
| Actor LR scale | 0.1                |

# Backup: Qualitative Depth Examples



Featureless terrain produces the weakest obstacle localization; structured textures preserve more usable geometry.